

CprE 308 Homework 1

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Student Name:

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Problem 1 (10 points)

Answer true (T) or false (F) for each following statements:

- ☐ 1. OS is responsible for managing the hardware resources in a computer.
- ☐ 2. Linux is one of the most widely used Microkernel OS.
- ☐ 3. The heap region of a process' address space is automatically allocated and deallocated.
- ☐ 4. System call is used by the hardware to communicate with the OS.
- ☐ 5. Context-switch time is overhead.
- ☐ 6. OS controls the execution of programs to prevent errors and improper use of the computer.
- ☐ 7. Monolithic kernel may have higher performance than Microkernel in practice because it incurs less communication overhead among modules.
- ☐ 8. Type 1 hypervisor requires support from an underlying OS.
- ☐ 9. The malloc() library call in C dynamically allocates memory in the stack region.

Problem 2 (5 points)

What is a process? What is a "Zombie" process?

Problem 3 (5 points)

Compare the advantage & disadvantage of Monolithic kernel and Microkernel.

Problem 4 (5 points)

What is the maximum depth of the stack for the following code in terms of number of stack frames (one stack frame per function call)? Be sure to count the first call to main.

```
int main() {  
    f(12);  
    return 1;  
}  
  
int f(int n) {  
    if (n <= 0)  
        return 0;  
    else  
        return f(n-1)+2*f(n-4);  
}
```

Problem 5 (5 points)

(1) How many processes does the following code create? (2) Draw the process tree of the program.

```
int main() {  
    int i;  
    for (i=1; i<3; i++)  
        fork();  
    return 1;  
}
```

Problem 6 (5 points)

Please write all possible outputs from the following piece of code:

```
int main(void) {  
    pid_t pid = fork();  
    if (pid > 0) {  
        printf("I am the parent\n");  
    } else if (pid == 0){  
        printf("I am the child\n");  
    }  
    else printf("ERROR!\n");  
    return 0;  
}
```

Problem 7 (5 points)

Consider the following code. Assume all system calls return successfully and the actual process IDs of the parent and child during the execution are 2600 and 2603, respectively. What are the values of pid/pid1 at lines A, B, C, D?

```
int main() {
    pid_t pid, pid1;
    pid = fork();
    if (pid < 0) {
        fprintf(stderr, "Fork Failed");
        return 1;
    } else if (pid == 0){
        pid1 = getpid();
        printf("child: pid = %d", pid); /* A */
        printf("child: pid1 = %d", pid1); /* B */
    }
    } else {
        pid1 = getpid();
        printf("parent: pid = %d", pid); /* C */
        printf("parent: pid1 = %d", pid1); /* D */
        wait(NULL);
    }
    return 0;
}
```